Personal details

Marcin Sztukowski

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LinkedIn
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Skills

Arduino
Bash
C
C#
C++
DataScience
Haskell
Java
Machine Learning
Networking
Python
Swift
SQL
Unix
Cryptography

Languages

English	C1
Polish	Native

Marcin Sztukowski

Projects

Jagiellonian University GMUM and Wawel Museum (Oct 2025 - Feb 2026)

2D→3D Reconstruction – VistaDream + Depth Anything

- Integrated Depth Anything v2 into VistaDream via custom adapter.
- Modified depth normalization and camera logic for stable 3D projection from paintings.
- Generated 3D Gaussian Splatting (.ply) point clouds from single 2D image.

ML Digital Painting Restoration – Diffusion + LoRA + Flux

- Built diffusion-based inpainting pipeline (FLUX.1-dev) for restoration of damaged artworks.
- Trained and optimized LoRA adapters for painter-specific style reconstruction.
- Improved reconstruction stability via mask refinement and controlled inference tuning.

FoxyGodot – 2D Platformer Game

- Developed 2D platformer in Godot (C#) with modular, object-oriented architecture.
- Implemented multiple design patterns: Strategy (enemy movement), State (player FSM), Factory (object creation), Observer/Event Bus (SignalManager), Singleton, Template Method, Iterator etc.
- Implemented centralized event system for player damage, score updates and game state transitions.
- Built extensible enemy framework with interchangeable movement strategies (walking, jumping, flying).

ECGApp – ECG Monitoring and ML Classification System

(iOS (SwiftUI) + Embedded BLE + Deep Learning)

Mobile Application (iOS)

- Developed iOS application from scratch using SwiftUI for real-time ECG visualization and recording
- Implemented BLE communication layer (CoreBluetooth) for streaming ECG data from external sensors
- Built configurable signal processing pipeline (sampling rate, scaling, normalization)
- Implemented local storage in .json and WFDB-compatible formats
- Deployed all ML models for on-device inference (CoreML) with full data privacy

Embedded & Hardware Integration

- Designed Arduino-based ECG acquisition system
- Implemented BLE data transmission to mobile device
- Built complete data flow: electrodes → microcontroller → Bluetooth → iOS app

Machine Learning and Signal Processing

- Trained three independent PyTorch models:
  - Rhythm classification
  - Beat-type classification
  - Waveform segmentation

- Performed preprocessing: resampling, wavelet denoising, normalization
- Converted models for mobile deployment

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Work experience

Seasonal Tax Office Assistant (Summer Intern) **Jul 2016 - Sep 2020**
Jand J FinancialAssociatesLLC,NewBritain,CT,USA

Summer employment each year in accounting and tax support roles. Tasks included invoice issuance, document proofreading, sales and property tax filing, and IT/technical support.

Warehouse Associate **Jun 2021 - Sep 2022**
TaniaKsiazkaSp. z o.o.,Bialystok

Fulfillment of online orders: picking, packing, inventory control. (two summer breaks)

Tax Office Assistant **Jun 2022 - Jan 2023**
Jand J FinancialAssociates LLC, New Britain , Connecticut, USA

Full-time assistant during tax season. Filing standard income tax returns and managing IRS and insurance audits.

Network Engineer **Dec 2024 - Jan 2025**
122MainSt,NewBritain, CT, USA

Complete overhaul of the building's IT and network infrastructure. Installation and configuration of routers, switches, patch panels, and IP surveillance cameras with new cabling. Configuration of VOIP server and handsets, VLAN segmentation, traffic routing, and allocation of distinct public IP addresses for each office.

IT Support / Lead Server Administrator **Jan 2022 - Present**
Jand J FinancialAssociatesLLC,NewBritain, CT, USA

Providing remote and telephone-based technical assistance to staff at a tax office. Maintaining seamless operation of all computing systems, VoIP phones, printers, email services, and the primary data server. Enhancing workflow efficiency through hardware/software upgrades and system optimizations for continuous, reliable operations.

Math Tutor **Jan 2022 - Present**

Tutored high school students in mathematics, focusing on core concepts and exam preparation.

Education

High school **Sep 2016 - May 2019**
VILO Białystok,

Economy - Bachelor's degree **Oct 2019 - Sep 2021**
University of Warsaw, Warsaw

Computer Science - Bachelor's degree **Oct 2021 - Jul 2025**
JagiellonianUniversity, Kraków

Computer Science - Masters's degree **Jul 2025 - Present**
JagiellonianUniversity, Kraków